How team-to-team coaching beats top-down curriculum

Why we gave up on writing a unified curriculum and started letting our cohorts teach each other.

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For most of our second year, three of us spent our evenings writing a curriculum. A real one. Sequenced units, learning objectives, mentor scripts, milestone projects. We had a spreadsheet with 240 rows in it. We had color coding. We were going to solve the consistency problem, finally, by writing down exactly what every cohort should learn and exactly when they should learn it, and then making sure every site followed the plan.

The curriculum is still on a shared drive somewhere. I have not opened it in fourteen months. The thing that actually solved the consistency problem turned out to be almost the opposite of what we built, and the path from one to the other is what this essay is about.

What top-down curriculum is actually good at

I want to be fair to the curriculum-first approach before I criticize it, because it does some things well and the people who design these documents are not wrong to do so. A well-written curriculum gives you a floor. It guarantees that a student who completes the program has been exposed to a defined set of ideas, in a defined order, regardless of which site they went to or which mentor they had. For a program that is trying to scale across many sites, or trying to certify outcomes to a funder, this is genuinely valuable. You can point to the document and say, this is what we teach. You can audit a session against the plan and identify whether it covered what it was supposed to cover. You can train a new mentor in a week by handing them the binder.

These are real benefits. I am not going to pretend they are not. The question is what they cost, and what the cost is hiding.

The cost is that the curriculum, no matter how carefully written, is a static document being applied to a dynamic situation. The students you have this Saturday are not the students the curriculum was written for. They are eleven specific kids, with eleven specific backgrounds, sitting in a specific room, on a specific day, after a specific week of school. The curriculum cannot know any of this. It can only assume an average student, and average students do not exist. Every kid in front of you is, on some axis or another, well off the average that the curriculum imagined. The mentor is then in the position of either following the plan and watching it land badly on most of the room, or deviating from the plan and undermining the consistency the plan was supposed to provide. Most mentors deviate, quietly, week after week, and the curriculum becomes a fiction that everyone agrees not to look at too closely.

This was where we were at the end of year two. We had a curriculum. The good mentors were ignoring it. The new mentors were following it and getting bad results. We had built exactly the wrong system: the people who needed structure least were free of it, and the people who needed support most were chained to a document that was actively making their sessions worse.

What we tried instead, mostly by accident

The change came from something that was not supposed to be a structural decision. We had a cohort in one of our sites that finished their twelve weeks ahead of schedule, mostly because their mentor was unusually strong and their group happened to gel well. We had another cohort, at a different site, that was three weeks behind, struggling with motivation and with a curriculum unit that wasn't landing. We did not have a good way to reassign mentors mid-season, so as a stopgap, we had two students from the ahead cohort come over and run a session with the behind cohort. They were thirteen and fourteen. They had been doing this themselves for ten weeks. They were not credentialed. We had nothing to lose.

The session went better than any of the previous three in that cohort had. Not slightly better. Visibly better. The students were more engaged, the questions they asked were sharper, and the thing they were stuck on, which had been a sensor calibration problem that the curriculum's suggested explanation was not landing on, got resolved in about twenty minutes. The two visiting students had been stuck on the same thing a month earlier, and they explained the fix the way they had figured it out, which was different from how the curriculum explained it, and which made sense to the room in a way the curriculum's version had not.

We did it again the next week with a different pair of visitors and got the same result. We started doing it deliberately. By the end of that season, about a third of our session-hours were being delivered, in some form, by other students. By the end of the next season, it was closer to half. The retention numbers went up. The unprompted-demo rates went up. The mentors, who you might have expected to feel sidelined, mostly reported the opposite, that their sessions were less exhausting and that the students were arriving more prepared.

Why students teach better than the document does

I want to be careful not to romanticize this. Students are not better than mentors at everything. There are things a thirteen-year-old cannot do that an adult mentor can, and we are not running a program where children replace the adults. The adults are still there, in every session, and the adults are still doing most of the actual coaching work. What changed is the source of the explanations.

There is a thing that happens when a student who learned something three weeks ago explains it to a student who is learning it now. The recent learner remembers what it felt like to not understand. They remember the specific moment of confusion, the specific bad mental model they had before the correct one clicked. The adult mentor learned this material five or ten or twenty years ago, and the bad mental model they had at the time, if they ever had one, is long gone. They explain the material from the standpoint of someone who finds it obvious. The recent learner explains it from the standpoint of someone who has just stopped finding it confusing, and that standpoint is much closer to where the new student is sitting.

There is also a thing that happens to the recent learner. Explaining something to another student forces a different kind of understanding than building it yourself does. The student who can make the robot follow a line is not necessarily the student who can tell you why it follows the line. The act of teaching the next cohort produces the second kind of understanding, and the second kind is the one that holds. We started noticing, after a year of this, that the students who had taught at least one session retained material dramatically better than the students who had only built. This was not a small effect. It was the largest single effect we could find on long-term retention of concepts.

The pedagogical literature on this is decades old and I am not going to pretend we discovered anything. The phrase "to teach is to learn twice" goes back to Joseph Joubert in the nineteenth century. What was new to us was how much of our actual program improvement, in measurable terms, came from leaning into this rather than treating teaching as a separate activity from learning.

What replaced the curriculum

We did not throw the curriculum away entirely. What we kept was a much shorter document, about a tenth the length of the original, that we now call the spine. The spine is a list of things every student should be able to demonstrate, in some form, by the end of their cohort. It is not sequenced. It is not scripted. It does not say what week each item should happen in. It says, here are the artifacts a student should have produced and the concepts they should be able to explain, by the time they leave us. The order, the pacing, and the method are up to the mentor and the cohort.

What replaced the sequenced curriculum is a library of session templates, mostly written by older students, that describe specific things that worked for them. A session template is not a script. It is closer to a recipe written by a friend, with the personal asides included. "When I was teaching the line follower, the thing that landed was telling them to imagine the robot as a person walking with their eyes closed and one hand on the wall. The book version with the proportional gain coefficient lost everyone. The hand-on-the-wall version, every kid got it in about ten minutes."

The library now has about 180 of these templates, across our curriculum spine. Mentors browse them before sessions. They are tagged by topic, by age range, and by what kind of student the template worked best for, in the original author's experience. New templates get added every season. Old templates that stop working get archived. The library is, in a real sense, the curriculum, but it is a curriculum that is being written by the students who lived through it, not by adults who imagined it from outside.

What it costs to run this way

I am not going to claim this model is easier than the top-down one. It is not. It is harder in several specific ways, and any program considering this approach should know what they are.

It is harder to onboard a new mentor. There is no binder you can hand them. You have to walk them through how to use the library, how to read a template, and how to write their own. This takes longer than reading a curriculum, and the payoff is delayed. A new mentor in this model is less effective in their first three sessions than a new mentor with a binder would be, and more effective by session ten. If your program does a lot of one-time volunteer engagement, this is a bad tradeoff. If your program has mentors who stay for at least a season, it is a good one.

It is harder to explain to funders. A funder wants to see the curriculum. They want to know what the kids are learning. Showing them a library of student-written session templates and a one-page spine is a harder pitch than showing them a 240-row spreadsheet. We have lost at least one grant over this, and I do not entirely blame the funder. The legibility cost is real.

It is harder to audit consistency across sites. The whole point of the top-down curriculum was that you could check whether a site was doing what it was supposed to. The team-to-team model gives up most of this. We audit on outputs now, not on process, and outputs are noisier than process. Some sites are stronger than others. We can see this in the demo rates. We cannot always immediately tell why, the way we could when we could just check whether the curriculum was being followed. We have had to develop different ways of identifying which sites need support, and we are still not great at this.

What you get for those costs is a program that adapts. The students are not the same year to year, the technology is not the same year to year, the mentors are not the same year to year, and a static document cannot keep up with any of this. A library of templates written by the people who actually ran the sessions can. After three years of running this model, the library is meaningfully better than the original curriculum was, because it has been pressure-tested by hundreds of sessions and revised by the people who saw what worked. The original curriculum was as good as three people writing in a coffee shop could make it. The library is as good as three years of cohorts could make it. There is no contest.

What this looks like from the outside

If you visited one of our sites today, you would not see a teacher at the front of the room. You would see an adult mentor sitting at a table with three students, and at the next table over, you would see a fifteen-year-old sitting with two ten-year-olds working through a sensor problem. The fifteen-year-old is not credentialed and is not pretending to be. She is using a session template she contributed to last summer, after she figured out the same problem herself, and the adults in the room can tell you that she is one of the most effective teachers in the program. She does not know that. We do not particularly tell her, because the moment you tell a fifteen-year-old that she is teaching, she starts performing teaching, and performing teaching is much worse than just doing what she was doing.

This is the program we have now. We did not design it. We stumbled into it because the document we had built did not work, and the thing that did work surprised us. If you are running a program and your curriculum is on its third revision and you are still not happy with how it lands in the room, I would suggest looking at what your most experienced students would say if you gave them a session to run. The answer might not be the program you planned. It might be the program you actually have.

Growing Up with Robotics runs cohort-based robotics programs across multiple sites in the DMV region. Our session template library is available to partner programs.